

Generative AI Integration

Generative AI Core

NLP Basics

- Text preprocessing
- Tokenization
- Stemming and Lemmatization
- Stop word removal
- Word embeddings
- Sequence modeling fundamentals
- Attention mechanism basics

Transformers & Large Language Models

- Transformer architecture
- Self-attention mechanism
- Encoder–Decoder structure
- Pretraining vs Fine-tuning
- Generative models vs discriminative models
- Context windows and inference
- Sampling techniques (temperature, top-k, top-p)

LLM APIs

- API-based LLM usage
- Model selection strategies
- Request/response structure
- Streaming responses
- Handling errors and retries

Prompt Engineering

- Prompt design principles

- System prompts
- User prompts
- Few-shot prompting
- Chain-of-thought prompting
- Role-based prompting
- Prompt evaluation and optimization

Function Calling / Tool Use

- Tool integration with LLMs
- Function calling concepts
- External API integration
- Structured output generation
- Tool orchestration

Token & Cost Management

- Tokenization concepts
- Token counting
- Context window management
- Cost optimization
- Rate limit handling

LangChain & LLM Frameworks

LangChain Fundamentals

- LangChain architecture
- Prompt templates
- LLM wrappers
- Chains
- Output parsers

Agents & Tools

- Agent architecture
- Tool integration
- Decision-making agents
- Multi-step reasoning workflows

Memory Systems

- Conversation memory
- Buffer memory
- Summary memory
- Persistent memory

Document Processing

- Document loaders
- File ingestion pipelines
- Handling structured and unstructured data

Text Splitting

- Chunking strategies
- Recursive text splitting
- Chunk size optimization
- Overlap strategies

LlamaIndex Basics

- Data connectors
- Index creation
- Query engines
- Retrieval pipelines

Vector Databases & RAG

Embeddings

- Vector representations
- Sentence embeddings
- Embedding models
- Similarity metrics

Vector Databases

- Vector indexing
- Similarity search
- Storage and retrieval
- Index optimization

Popular Vector Databases

- FAISS

- Pinecone
- Chroma
- Weaviate

Retrieval Augmented Generation (RAG)

- RAG architecture
- Data ingestion pipelines
- Retrieval workflows
- Context injection

Semantic Search

- Meaning-based search
- Similarity ranking
- Query embeddings
- Search optimization

Hybrid Search

- Keyword search integration
- Vector + keyword ranking
- Hybrid retrieval pipelines

Generative AI Use Case Implementations

AI Chatbot Development

- Conversational interface design
- Backend integration
- Memory-enabled chatbots
- Web-based deployment

Resume Screening System

- Resume parsing
- Embedding generation
- Job description matching
- Candidate ranking system

AI-Powered Search Engine

- Knowledge base creation
- Document indexing

- Semantic retrieval
- Answer generation

Code Generation Assistant

- Programming prompt design
- Code generation workflows
- Code explanation
- Debugging assistance

Content Generation Platform

- Blog/article generation
- Marketing copy generation
- Social media content creation
- Template-based generation

Question Answering on Documents

- Document ingestion
- Vector indexing
- Query processing
- Context-based answer generation

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